

ACTIVITY 2A

MATLAB / SCILAB FUNCTIONS

1. Implement a function that will translate the point (x, y) by a specified shift along the x-axis and y-axis which is the (x', y') . The (x', y') is defined as:
 - New coordinate: $x' = x + dx$
 - New coordinate: $y' = y + dy$

Provide at least 3 examples of inputs in the given problem.

2. Create a function to find the area of the shaded portion and the unshaded portion where the input parameter is the side of the square.



3. Create a MATLAB function to calculate the mean of a set of numbers, where the input parameter is a vector containing exactly 20 elements. If the input does not contain exactly 20 numbers, the function should automatically return the message: "Input must be a vector containing exactly 20 numbers".